The Art of Solving Thermodynamics Problems

How to Plan Your Attack

The five buckets

Solving every closed-system problem for an Ideal Gas relies on understanding the five categories of equations available:

#	Bucket	What is in it
1	Definitions	$\rho = 1/v, \quad h = u + Pv, \quad c_v = (\frac{\partial u}{\partial T})_v, \quad c_p = (\frac{\partial h}{\partial T})_P$
2	Equations of State	(Ideal Gas) $\quad Pv = RT, \quad u = c_v T$
3	Conservation	$\Delta u = q - w$
4	Work & heat	$w_b = \int P\, dv \quad$ (q from rearranging Conservation)
5	Process spec	iso-v / iso-P / iso-T / iso-s (quasistatic adiabatic) $\Rightarrow \; P(v)$

Thermo problems = puzzles

- Know the **rules** (equations from buckets)
- Identify **given** information
- Identify **unknown** information
- Use rules to **deduce** unknowns from givens, progressively
- Game: **Assemble equations until you have enough equations for all unknowns**

The starting point: fixing a state

To calculate anything at a state, you must first know **two independent intensive properties** there. *(the State Postulate)*

- For an **Ideal Gas**, that means **any two of** P**,** v**,** T.
- The **third** then follows from the equation of state, $Pv = RT$.

- Internal energy and enthalpy depend on T **alone**:

$$u = c_v T, \qquad h = c_p T.$$

- So getting T is what unlocks u and h.

So the whole game: state to state

- At each state, spot the **two** known of P, v, T; the EoS gives the **third**.
- The **process** to the next state hands you one property there, so you can fix it too.
- Move along, state by state, until P**,** v**,** T **are known everywhere**.
- *Only then* do we work out the energy changes, work, and heat.

Our example

We will plan and solve this one together, step by step

A typical thermo problem statement

Air in a pneumatic cylinder starts at 100 kPa, 300 K. It is compressed isothermally to $v = 0.2\ \mathrm{m}^3/\mathrm{kg}$*, then cooled at constant volume until 250 K. Find the net work output and the net heat input.*

- Dense to read
- Where do we start?!
- How do we proceed further?

Step 0 – every time!

- We will always work with Perfect Ideal Gases.
- **Which Ideal Gas is it?**
- Put down from Table A-2:

$$R,\ c_v,\ c_p = c_v + R,\ k = c_p/c_v.$$

Gas	R	c_p	c_v	k
Air	0.287	1.005	0.718	1.40
N_2	0.297	1.039	0.743	1.40
O_2	0.260	0.918	0.658	1.40
CO_2	0.189	0.846	0.657	1.29

- *from Table A-2 in the textbook*
- units for R, c_p and c_v kJ/(kg·K)
- k is a dimensionless ratio

Step 0 for our example

Air in a **pneumatic cylinder** starts at 100 kPa, 300 K. It is compressed isothermally to $v = 0.2\ \mathrm{m}^3/\mathrm{kg}$, then cooled at constant volume until 250 K. Find the net work output and the net heat input.

From Table A-2 for **air**:

- $R = 0.287$ kJ/(kg·K)
- $c_v = 0.718$ kJ/(kg·K)
- $c_p = 1.005$ kJ/(kg·K)
- $k = 1.4$

Also note "pneumatic cylinder" is telling us that this is a closed system.

Step 1: Key States and processes connecting them

- Find the **key states**; number them and count them.
- Identify the **canonical process** between states
- Each process is one of the following:
    - constant-volume or isochoric: iso-v
    - constant-pressure or isobaric: iso-P
    - constant-temperature or isothermal: iso-T
    - quasistatic adiabatic or isentropic: iso-s

Step 1 for our example

Colour key: state information · process information

Step 2: Plan with an arrow diagram

- Lay the states left to right.
- Write P-v-T **known**, and unknowns as **“?”**.
- Put the process and its $P(v)$ under each arrow.
- Write any given q or w under each arrow.

Step 2 for our example

State 1	State 2	State 3
P1 = 100 kPa	v2 = 0.2	T3 = 250 K
T1 = 300 K	P2 = ?	v3 = ?
v1 = ?	T2 = ?	P3 = ?

1 → 2 → 3

isothermal (T const)

Δu = ? w= ? q = ?

isochoric (v const)

Δu = ? w= ? q = ?

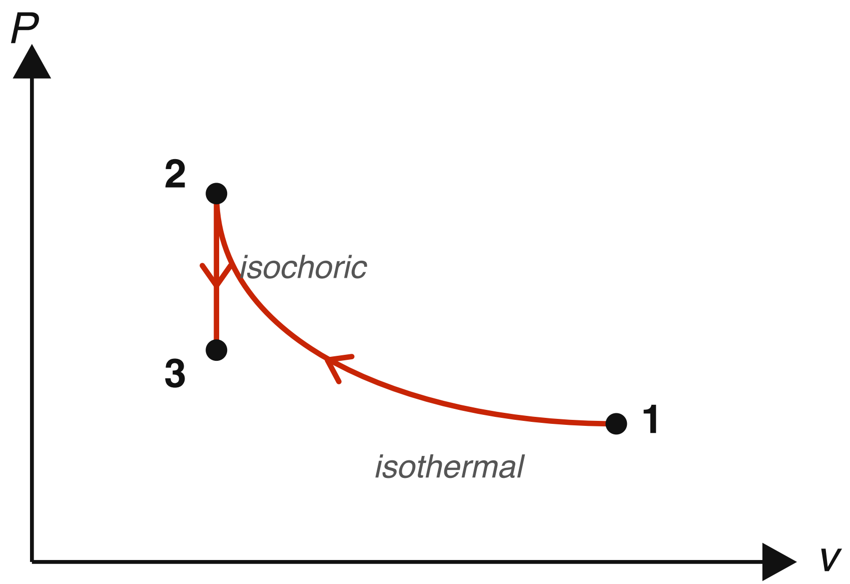
Air (ideal gas). Find the net work and net heat. (Pressures in kPa, v in m^3/kg.)

Step 3: sketch the $P - v$ diagram

- The arrow diagram fixes the **topology**.
- The $P - v$ sketch visualizes the **shape** of each path.
- It is your check that the processes join up sensibly.
- And, work for each process is area under the curve.

Step 3 for our example

- $1 \to 2$ isothermal: a curve down to smaller v.
- $2 \to 3$ isochoric: straight down at constant v.
- Areas under the curves are the works w_b.



Step 4: Stage 1 – fix (P, v, T) of every state

- Start at the state with **two** of P, v, T; the EoS gives the third.
- For each next state: **given** supplies one, the **process** supplies the second, the EoS gives the third.
- Don't compute yet – just **write the equation** under each state.

Step 4: Stage I for our example

State 1	State 2	State 3
P1 = 100 (given)	v2 = 0.2 (given)	T3 = 250 (given)
T1 = 300 (given)	T2 = T1 (process)	v3 = v2 (process)
v1 = R T1 / P1 (EoS)	P2 = R T2 / v2 (EoS)	P3 = R T3 / v3 (EoS)

$$1 \xrightarrow{\text{isothermal}} 2 \xrightarrow{\text{isochoric}} 3$$

Two knowns at each state → the Ideal-Gas EoS gives the third.

Step 4: Stage 2 – work and heat, process by process

For each process, three equations:

$$\Delta u = c_v \, \Delta T$$

$$w_b = \int P \, dv$$

$$q = \Delta u + w_b$$

The work integral is evaluated along that process's $P(v)$.

Step 4: Stage 2 – work for each process

P-v-T are fixed now (Stage 1), so Δu, w_b and q follow from them.

Denoting the start and end states as 1 and 2:

$$\Delta u = c_v \left(T_2 - T_1\right)$$

iso-v	**iso-P**	**iso-T**	**iso-s**
$dv = 0$	$P = P_1 = P_2$	$P = RT_1/v = RT_2/v$	$P = P_1 v_1^k/v^k = P_2 v_2^k/v^k$
$w_b = \int_1^2 P\,dv$	$w_b = \int_1^2 P\,dv$	$w_b = \int_1^2 P\,dv$	$w_b = \int_1^2 P\,dv$
$= 0$	$= P_1 \left(v_2 - v_1\right)$	$= RT_1 \ln \frac{v_2}{v_1}$	$= c_v \left(T_1 - T_2\right)$

then

$$q = \Delta u + w_b$$

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Step 4: Stage 2 for our example

State 1

P1 = 100 (given)
T1 = 300 (given)
v1 = R T1 / P1 (EoS)

State 2

v2 = 0.2 (given)
T2 = T1 (process)
P2 = R T2 / v2 (EoS)

State 3

T3 = 250 (given)
v3 = v2 (process)
P3 = R T3 / v3 (EoS)

$$1 \xrightarrow{\text{isothermal}} 2 \xrightarrow{\text{isochoric}} 3$$

isothermal

$\Delta u = cv\ (T2 - T1)$
$w = R\ T1\ \ln(v2 / v1)$
$q = \Delta u + w$

isochoric

$\Delta u = cv\ (T3 - T2)$
$w = 0$
$q = \Delta u + w$

The entire calculation plan for the problem.

Step 5 – finish

- Add with their signs:
  - **net work** $= \sum w_b$
  - **net heat** $= \sum q$
- Then anything further required:
  - efficiency, mass required, and so on.

Step 5 for our example

State 1	State 2	State 3
P1 = 100 (given)	v2 = 0.2 (given)	T3 = 250 (given)
T1 = 300 (given)	T2 = T1 (process)	v3 = v2 (process)
v1 = R T1 / P1 (EoS)	P2 = R T2 / v2 (EoS)	P3 = R T3 / v3 (EoS)

$$1 \xrightarrow{\text{isothermal}} 2 \xrightarrow{\text{isochoric}} 3$$

isothermal

$\Delta u = cv\ (T2 - T1)$

$w = R\ T1\ \ln(v2 / v1)$

$q = \Delta u + w$

isochoric

$\Delta u = cv\ (T3 - T2)$

$w = 0$

$q = \Delta u + w$

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net work = w(1→2) + w(2→3)

net heat = q(1→2) + q(2→3)

The plan – in one line

Which IG $\rightarrow$ states + processes $\rightarrow$ arrow diagram $\rightarrow$ $P - v$ sketch $\rightarrow$ $P - v - T$ $\rightarrow$ Δu, w_b, q $\rightarrow$ finish

You can now attack any PIG problem

- Same steps for engines, pneumatics, cycles.
- The arrow-diagram gives you the roadmap.
- Follow the map for subsequent calculations.